Jiechao Gao

CONTACT INFORMATION

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Scholar
Homepage

Y2E2
Center for Sustainable Development & Global Competitiveness
Stanford University
Stanford Profile
Stanford, CA

STATUS

On the 2026–27 academic job market. U.S. Permanent Resident, no visa sponsorship required.

RESEARCH INTERESTS

I work on building **reliable AI systems**. My research follows three directions:

- **Trustworthy Generation & Agents** **(NeurIPS'26 ×5, ICML'26 ×2, ICLR'26, AAAI'26 ×2, WWW'26, ACL'26, Findings of ACL'26, etc.)**
- **Alignment at Large Scale** **(NeurIPS'25, SIGIR'26, KDD'25 ×2, ICML'25, ACM MM'25 ×2, etc.)**
- **Privacy-Preserving & Efficient Learning** **(NPJ AI'26, ICCPS'25, SenSys, IEEE TSC'20, IPDPS'20, EMNLP'25, etc.)**

My long-term goal is **Translational AI Systems**, reliable AI for real-world domains such as healthcare, buildings, finance, software and recommendation, etc. I build and test this work with clinicians, building engineers, developers and platform teams. Parts of this work now run in open source (FinRL, Harbor) and in a recommendation system used by hundreds of millions of people.

PROFESSIONAL EXPERIENCE

Stanford University
Postdoc Research Associate — Jul. 2025 - now

- Research on trustworthy generative models and resource-efficient LLM systems, with applications in engineering and sustainability.
- **Advisors**: Prof. Jie Wang and Prof. Michael Lepech

Columbia University
Associate Research Scientist — Aug. 2024 - June 2025

- Federated learning, reinforcement learning, and trustworthy LLM agents for financial applications; core contributor and contest organizer of the FinRL open-source ecosystem.

INDUSTRY EXPERIENCE

Frees Fund (VC), Beijing
AI Investment Intern — May 2018 - Aug. 2018

EDUCATION

Ph.D. in Computer Science — Aug. 2018 - Aug. 2024
University of Virginia — Charlottesville, VA
Department of Computer Science

- **Advisor**: Prof. Bradford Campbell

Master of Science in Electrical Engineering — Sep. 2016 - Feb. 2018
Columbia University — New York, NY
Department of Electrical Engineering

B.S. in Applied Physics & B.Eng. in Financial Engineering — Sep. 2012 - Jul. 2016
Jilin University — Changchun, Jilin
Department of Physics & Department of Economics

Awards

Google HE Faculty AI Fellowship	2026
EMNLP Industry Track Best Paper Nomination	2025
World's Top 2% Scientists (Stanford/Elsevier)	2024–2026
ACM BuildSys Best Paper Candidate	2024
ICPP NSF Travel Award	2023
UVA Link Lab Flash Talk Awardee	Feb. 2022
Excellent Graduation Thesis (5/242), JLU Physics Department	Jun. 2016
MCM Outstanding Winner Award	Apr. 2015
National Scholarship (Top 1%)	2013 & 2014
Manulife Financial Center Outstanding Intern Award	2014

Publications

Scholar

4,758 Citations · H-index 29 · i10-index 56 (Google Scholar, as of Sep. 2026)

111 peer-reviewed publications, with **70 as first / co-first / corresponding author**. Venues include **NeurIPS ×7, ICML ×3, ICLR, AAAI ×2, KDD ×3, WWW, SIGIR, ACM MM ×3, CVPR, the ACL family ×12 (ACL / EMNLP / EACL, incl. Industry & Findings), NPJ Artificial Intelligence, ICCPS, CHIL, IPDPS, and IEEE TSC**; with 1 Best Paper Nomination (EMNLP Industry '25), 1 Best Paper Candidate (BuildSys '24), and 2 Orals (ACM MM '25 top 4%; CHIL '23 13.3%).

∗ equal contribution † corresponding author

Peer-reviewed Conference Papers (with proceedings)

NeurIPS '26 67. **Jiechao Gao**, Chang Liu, Yuandong Pan, Ying Liu, Michael Lepech
Anchoring LLM-based Chest X-ray Report Generation via Diffusion Language Planning
Conference on Neural Information Processing Systems, 2026

NeurIPS '26 66. Haoran Ma*, **Jiechao Gao***, Shisong Tang, Bing Han, Jifeng Hu, Hechang Chen
Render Structure Uncertainty for HTML Repair in MLLM-based UI-to-Code Generation
Conference on Neural Information Processing Systems, 2026

NeurIPS '26 65. Haoran Ma, Linxiao Li, Chenyue Wang, Haochen Sui, **Jiechao Gao**†
Phase-wise MLLM Tuning for Multi-framework WebUI Code Generation
Conference on Neural Information Processing Systems, 2026

NeurIPS '26 64. Linxiao Li, Haoran Ma, Chenyue Wang, Jiaye Lin, Haochen Sui, **Jiechao Gao**†
Toward Executable Multi-framework Front-end Code Generation with Self-Correction
Conference on Neural Information Processing Systems, 2026

NeurIPS D&B '26 63. Lin Shi, . . . , **Jiechao Gao**, . . . , Alex Shaw
Harbor Adapters and Harbor-Index: Infrastructure and a Curated Meta-Dataset for Large-Scale Agentic Evaluation
Conference on Neural Information Processing Systems, Evaluations and Datasets Track, 2026

ICML '26 62. Zixiong Yang*, Linxiao Li*, Jiaye Lin, Binrui Wu, Xiaoyu Kang, **Jiechao Gao**†
Deterministic Component Mining for Multi-framework UI2Code Generation
International Conference on Machine Learning, 2026

ICML '26 61. Jie Wu*, Haoran Ma*, Shisong Tang, Yulin Xu, Xiaoyu Kang, **Jiechao Gao**†
MulFCoder: Framework-conditioned Multi-agent for MLLM-based Multi-framework Front-end Code Generation
International Conference on Machine Learning, 2026

ICLR '26 60. Siyuan Chen*, Hang Ding*, Kangxiao Yu, **Jiechao Gao**†
GRO-RAG: Gradient-aware Re-rank Optimization for Multi-source Retrieval-Augmented Generation
International Conference on Learning Representations, 2026

AAAI '26 59. **Jiechao Gao**†, Chang Liu, Yuangang Li
S2D-ALIGN: Shallow-to-Deep Auxiliary Learning for Anatomically-Grounded Radiology Report Generation
AAAI Conference on Artificial Intelligence, 2026

AAAI '26
58. Yuangang Li*, Yiqing Shen*, Yi Nian, **Jiechao Gao**, Ziyi Wang, Chenxiao Yu, Shawn Li, Jie Wang, Xiyang Hu†, Yue Zhao†
Mitigating hallucinations in large language models via causal reasoning
AAAI Conference on Artificial Intelligence, 2026

KDD '26 Blue Sky
57. Yixiang Yao, Yuhang Yao, Xinyi Fan, **Jiechao Gao**, Jie Wang, Minjia Zhang, Srivatsan Ravi, Carlee Joe-Wong
Trustworthy Agent Network: Trust in Agent Networks Must Be Baked In, Not Bolted On
ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Blue Sky Track, 2026

WWW '26
56. Sahil Tripathi, Gautam Siddharth Kashyap, Mehwish Nasim, Jian Yang, **Jiechao Gao**†, Usman Naseem
They Said Memes Were Harmless-We Found the Ones That Hurt: Decoding Jokes, Symbols, and Cultural References
The ACM Web Conference, 2026

SIGIR '26
55. Qi Sun, Yulin Xu, Zelin Wang, Xiaoyu Kang, Keyan Jin, **Jiechao Gao**†
Distribution-aware Re-representations for Multi-Scenario Recommendations
International ACM SIGIR Conference on Research and Development in Information Retrieval, 2026

ACL '26 Ind.
54. Abdullah Mohammad, Sushant Kumar Ray, Pushkar Arora, Rafiq Ali, Ebad Shabbir, Gautam Siddharth Kashyap, **Jiechao Gao**†, Usman Naseem†
Are Large Language Models Economically Viable for Industry Deployment?
Annual Meeting of the Association for Computational Linguistics, Industry Track, 2026

ACL '26 Ind.
53. Soumedhik Bharati*, Ebad Shabbir*, **Jiechao Gao**†
FROST: Factual Reasoning via Optimized Stochastic Trajectories in Large Language Models during Inference
Annual Meeting of the Association for Computational Linguistics, Industry Track, 2026

ACL Findings '26
52. **Jiechao Gao**†, Rohan Kumar Yadav, Yuangang Li, Yuandong Pan, Jie Wang, Ying Liu, Michael Lepech
LLM-Guided Semantic Bootstrapping for Interpretable Text Classification with Tsetlin Machines
Findings of the Association for Computational Linguistics: ACL 2026

EMNLP '26 Ind.
51. Muran Yu, **Jiechao Gao**, Yuandong Pan, Barney Haoyun Miao, Andrew Carter Lesh, Kincho Law, Jie Wang, Michael Lepech
EGT-KG: Evidence-Grounded Typed KG Retrieval for Practical Scientific QA with Small Language Models
Conference on Empirical Methods in Natural Language Processing, Industry Track, 2026

EACL '26 Ind.
50. Sushant Kumar Ray, Gautam Siddharth Kashyap, Sahil Tripathi, Nipun Joshi, Vijay Govindarajan, Rafiq Ali, **Jiechao Gao**†, Usman Naseem
Do Clinical Question Answering Systems Really Need Specialised Medical Fine Tuning?
Conference of the European Chapter of the Association for Computational Linguistics, Industry Track, 2026

EACL '26 Find.
49. Gautam Siddharth Kashyap, Harsh Joshi, Niharika Jain, Ebad Shabbir, **Jiechao Gao**†, Nipun Joshi, Usman Naseem†
Revealing the Truth with ConLLM for Detecting Multi-Modal Deepfakes
Conference of the European Chapter of the Association for Computational Linguistics, 2026

ICASSP '26
48. Ebad Shabbir*, Pushkar Arora*, Rakshita Saksainaa, Tiange Xie, **Jiechao Gao**†
AVATAR: Audio-Visual Adaptive Fusion via Trained Agent Reinforcement for Multimodal Deepfake Detection
IEEE International Conference on Acoustics, Speech and Signal Processing, 2026

ICASSP '26
47. Yuheng Li*, Taoyu Wu*, **Jiechao Gao**†, Wei Zhu†, Wenwen Ouyang
DemoReranker: Enhancing the In-Context Learning Capability of Multi-Modal Large Models via Demo Ranking
IEEE International Conference on Acoustics, Speech and Signal Processing, 2026

ICASSP '26
46. Xinyuan Huang, **Jiechao Gao**†, Tiange Xie
Differentiable Meta-Optimization for Federated Neural Architecture Search (MO-FedNAS)
IEEE International Conference on Acoustics, Speech and Signal Processing, 2026

ICASSP '26
45. Taoyu Wu, Yuheng Li, **Jiechao Gao**†, Wei Zhu†, Ming Tan, Hui Yi Leong
EPO: Enhanced Preference Optimization with Multi-Response Data for LLMs via Stochastic Softmax
IEEE International Conference on Acoustics, Speech and Signal Processing, 2026

ICML '25
44. Shisong Tang*, Chao Cui*, Fan Li, **Jiechao Gao**, Hechang Chen
Calibrating Video Watch-time Predictions with Credible Prototype Alignment
International Conference on Machine Learning, 2025

NeurIPS '25
43. Yidong Wu*, Siyuan Chen*, Binrui Wu, Fan Li, **Jiechao Gao**†
Adaptive Gradient Masking for Balancing ID and MLLM-based Representations in Recommendation
Conference on Neural Information Processing Systems, 2025

KDD '25
42. Yulin Xu*, Chao Cui*, Shisong Tang*, Fan Li, Bing Han, Huafeng Cao, **Jiechao Gao**†, Hechang Chen
Leveraging Label Distributions as Anchors to Enhance Video Recommendation
ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2025

KDD '25
41. Binrui Wu*, Shisong Tang*, Fan Li, Bing Han, Chang Meng, Jingyu Xiao, **Jiechao Gao**†
Aligning and Balancing ID and Multimodal Representations for Recommendation
ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2025

ACL Findings '25
40. Xunzhu Tang*, **Jiechao Gao***, Jin Xu*, Tiezhu Sun, Yewei Song, Saad Ezzini, Wendkûuni C. Ouédraogo, Jacques Klein, Tegawendé F. Bissyandé
SynFix: Dependency-Aware Program Repair via RelationGraph Analysis
Findings of the Association for Computational Linguistics: ACL 2025

EMNLP '25 Ind.
Best Paper Nom.
39. Mohammad Anas Azeez*, Rafiq Ali*, Ebad Shabbir, Zohaib Hasan Siddiqui, Gautam Siddharth Kashyap, **Jiechao Gao**†, Usman Naseem†
Truth, Trust, and Trouble: Medical AI on the Edge
Conference on Empirical Methods in Natural Language Processing, Industry Track, 2025

EMNLP '25 Ind.
38. Zohaib Hasan Siddiqui*, **Jiechao Gao***, Ebad Shabbir*, Mohammad Anas Azeez, Rafiq Ali, Gautam Siddharth Kashyap, Usman Naseem†
LLMs on a Budget? Say HOLA
Conference on Empirical Methods in Natural Language Processing, Industry Track, 2025

EMNLP '25 Ind.
37. Hui Yi Leong, Yuheng Li, Yuqing Wu, Wenwen Ouyang, Wei Zhu†, **Jiechao Gao**†, Wei Han
AMAS: Adaptively Determining Communication Topology for LLM-based Multi-agent System
Conference on Empirical Methods in Natural Language Processing, Industry Track, 2025

EMNLP '25 Ind.
36. Yuheng Li*, **Jiechao Gao***, Wei Han, Wenwen Ouyang, Wei Zhu†, Hui Yi Leong
FT-MDT: Extracting Decision Trees from Medical Texts via a Novel Low-rank Adaptation Method
Conference on Empirical Methods in Natural Language Processing, Industry Track, 2025

IJCNLP-AACL '25
35. Sahil Tripathi, MD Tabrez Nafis, Imran Hussain, **Jiechao Gao**†
The Confidence Paradox: Can LLM Know When It Is Wrong?
International Joint Conference on Natural Language Processing and AACL, 2025

ACM MM '25
Oral (top 4%)
34. Binrui Wu*, Haochen Sui*, Jiaye Lin*, **Jiechao Gao**†, Ting Xu, Keyan Jin, Xuesong Zhang
Prototype-Guided Representation Projection for Multi-Domain Multi-Task Recommendation
ACM International Conference on Multimedia, 2025

ACM MM '25
33. Songning Lai*, Ninghui Feng*, **Jiechao Gao***, Hao Wang, Haochen Sui, Xin Zou, Jiayu Yang, Wenshuo Chen, Lijie Hu, Hang Zhao, Xuming Hu, Yutao Yue
From Guesswork to Guarantee: Towards Faithful Multimedia Web Forecasting with TimeSieve
ACM International Conference on Multimedia, 2025

CVPR '25
32. Yuzhi Huang*, Chenxin Li*†, Haitao Zhang, Zixu Lin, Yunlong Lin, Hengyu Liu, Wuyang Li, Xinyu Liu, **Jiechao Gao**, Yue Huang†, Xinghao Ding, Yixuan Yuan
Track Any Anomalous Object:A Granular Video Anomaly Detection Pipeline
IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025

ICCPS '25
31. **Jiechao Gao***, Mingyue Tang*, Wenpeng Wang, Tushar Routh, Brad Campbell
Atlas: Ensuring Accuracy for Privacy-Preserving Federated IoT Applications
ACM/IEEE International Conference on Cyber-Physical Systems, 2025

ICASSP '25
30. Gautam Kashyap, Zohaib Hasan Siddiqui, Mohammad Anas Azeez, Rafiq Ali, Shantanu Kumar, Navin Kamuni, **Jiechao Gao**†
Fooling the Forgers: A Multi-Stage Framework for Audio Deepfake Detection
IEEE International Conference on Acoustics, Speech and Signal Processing, 2025

ICASSP '25 29. Gautam Kashyap, Harsh Joshi, Manaswi Kulahara, Rajkumar Dhakar, Atul Sajjanhar, **Jiechao Gao**†, Sarthak Jain, Shahab Sohail
Can AI See What We Can't? Leveraging Deep Learning and Multi-Temporal Satellite Data to Revolutionize Crop Type Mapping and Yield Prediction
IEEE International Conference on Acoustics, Speech and Signal Processing, 2025

BigData '25 28. **Jiechao Gao**, Rohan Kumar Yadav, Xinyuan Huang, Jie Wang
Enhancing Interpretability in Self-Training with Tsetlin Machines for Mitigating Noisy Pseudo-Labels
IEEE International Conference on Big Data, 2025

BigData '25 27. Xinyuan Huang, **Jiechao Gao**†, Jie Wang
Federated Neural Architecture Search with Model-Agnostic Meta Learning
IEEE International Conference on Big Data, 2025

BIBM '25 26. Xinyuan Huang, **Jiechao Gao**†
Toward Fair and Efficient Neural Architecture Search in Heterogeneous Federated Learning
IEEE International Conference on Bioinformatics and Biomedicine, 2025

BIBM '25 25. Xinyuan Huang, Hikaru Kurosawa, Jack Wunder, Sujit Patil, **Jiechao Gao**, Karthik Kuber, Jonathan C. Irish, Michael J. Daly
U-KAN: Hybrid Spatial-Functional Deep Learning for Tumor Depth Estimation in Fluorescence-Guided Cancer Surgery
IEEE International Conference on Bioinformatics and Biomedicine, 2025

KSEM '25 24. Huaibin Ye, Zuobin Ying, **Jiechao Gao**, Ximeng Liu, Jianping Cai
FedBCE: Rethinking Clustered Federated Learning for Better Clustering Efficiency
International Conference on Knowledge Science, Engineering and Management, 2025

ASONAM '25 23. Arpita Soni, Sahil Tripathi, Gautam Siddharth Kashyap, Manaswi Kulahara, Mohammad Anas Azeez, Zohaib Hasan Siddiqui, Nipun Joshi, **Jiechao Gao**†
Can We Predict Your Next Move Without Breaking Your Privacy?
IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining, 2025

SEC '25 22. Marshall Clyburn*, Nabeel Nasir*, Md Fazlay Rabbi Masum Billah, Victor Ariel Leal Sobral, **Jiechao Gao**, Fateme Nikseresht, Brad Campbell
fReeLoaders: An IoT Ecosystem for Real-Time Deadline-Driven Task Scheduling using Reinforcement Learning
ACM/IEEE Symposium on Edge Computing, 2025

ACM MM '24 21. Yiyong Xiao*, Kai Shu*, Haoyi Zhang*, Baohua Yin, Wai Seng Cheang, Haoyang Wang, **Jiechao Gao**†
EGGesture: Entropy-Guided Vector Quantized Variational AutoEncoder for Co-Speech Gesture Generation
ACM International Conference on Multimedia, 2024

BuildSys '24 20. Tushar Routh, Nurani Saoda, Fateme Nikseresht, Md Fazlay Rabbi Masum Billah, **Jiechao Gao**, Viswajith Govinda Rajan, Bradford Campbell
Best Paper Cand.
ScreenSense: Screen Activity Detection in Real-World Environments with Indoor Light Sensors
ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, 2024

BigData '24 19. Jiayu Wang, **Jiechao Gao**†
Federated Learning with Knowledge Distillation to Mitigate Catastrophic Forgetting and Data Heterogeneity in IoV Systems
IEEE International Conference on Big Data, 2024

BIBM '24 18. **Jiechao Gao**†, Yuangang Li
FedMetaMed: Federated Meta-Learning for Personalized Medication in Distributed Healthcare Systems
IEEE International Conference on Bioinformatics and Biomedicine, 2024

BIBM '24 17. Kai Shu*, Yuzhuo Jia*, Ziyang Zhang, **Jiechao Gao**†
FODA-PG for Enhanced Medical Imaging Narrative Generation: Adaptive Differentiation of Normal and Abnormal Attributes
IEEE International Conference on Bioinformatics and Biomedicine, 2024

SciSec '24 16. Tiange Xie, Feng Liu, **Jiechao Gao**, Yinghui Wang
Integrating Consortium Blockchain and Attribute-Based Searchable Encryption for Automotive Threat Intelligence Sharing Model
International Conference on Science of Cyber Security, 2024

ICDM SSTDM '24
15. **Jiechao Gao**†, Yuangang Li, Syeda Faiza Ahmed
Fed-LDR: Federated Local Data-infused Graph Creation with Node-centric Model Refinement
IEEE ICDM International Workshop on Spatial and Spatiotemporal Data Mining, 2024

CHIL '23
Oral (13.3%)
14. Mingyu Tang*, **Jiechao Gao***, Guimin Dong, Carl Yang, Bradford Campbell, Brendan Bowman, Jamie Marie Zoellner, Emaad Abdel-Rahman, Mehdi Boukhechba
SRDA: Mobile Sensing based Fluid Overload Detection for End Stage Kidney Disease Patients using Sensor Relation Dual Autoencoder
PMLR Conference on Health, Inference, and Learning, 2023

ICPP '23
13. **Jiechao Gao**, Wenpeng Wang, Fateme Nikseresht, Viswajith Govinda Rajan, Bradford Campbell
PFDRL: Personalized Federated Deep Reinforcement Learning for Residential Energy Management
International Conference on Parallel Processing, 2023

DCOSS '23
12. Wenpeng Wang*, Fateme Nikseresht*, Viswajith Govinda Rajan, **Jiechao Gao**, Bradford Campbell
Enabling ubiquitous occupancy detection in smart buildings: A wifi ftm-based approach
International Conference on Distributed Computing in Smart Systems and the Internet of Things, 2023

PRCV '23
11. Wei Cai, **Jiechao Gao**
SSCL: semi-supervised contrastive learning for industrial anomaly detection
Chinese Conference on Pattern Recognition and Computer Vision, 2023

NeurIPS D&B '22
10. Xiao-Yang Liu*, Ziyi Xia*, Jingyang Rui, **Jiechao Gao**, Hongyang Yang, Ming Zhu, Christina Dan Wang†, Zhaoran Wang, Jian Guo†
FinRL-Meta: Market Environments and Benchmarks for Data-Driven Financial Reinforcement Learning
Conference on Neural Information Processing Systems, Datasets and Benchmarks Track, 2022

COMSNETS '22
9. Priyan Malarvizhi Kumar, Bharat Rawal, **Jiechao Gao**
Blockchain-enabled privacy preserving of IoT data for sustainable smart cities using machine learning
International Conference on COMmunication Systems & NETworkS, 2022

IPSN '21
8. Md Fazlay Rabbi Masum Billah, Nurani Saoda, **Jiechao Gao**, Bradford Campbell
BLE can see: a reinforcement learning approach for RF-based indoor occupancy detection
ACM/IEEE International Conference on Information Processing in Sensor Networks, 2021

BuildSys '21
7. Tong Wu*, Murtadha Aldeer*, Tahiya Chowdhury, Amber Haynes, Fateme Nikseresht, Mahsa Pahlavikhah Varnosfaderani, **Jiechao Gao**, Arsalan Heydarian, Brad Campbell, Jorge Ortiz
The smart building privacy challenge
ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, 2021

BuildSys '21
6. Wenpeng Wang, Zetian Liu, **Jiechao Gao**, Nurani Saoda, Bradford Campbell
UbiTrack: Enabling scalable & low-cost device localization with onboard wifi
ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, 2021

ICPP '21
5. Haoyu Wang, Haiying Shen, **Jiechao Gao**, Kevin Zheng, Xiaoying Li
Multi-agent reinforcement learning based distributed renewable energy matching for datacenters
International Conference on Parallel Processing, 2021

ICAIF '21
4. Xiao-Yang Liu, Hongyang Yang, **Jiechao Gao**, Christina Wang
FinRL: Deep Reinforcement Learning Framework to Automate Trading in Quantitative Finance
ACM International Conference on AI in Finance, 2021

IPDPS '20
3. **Jiechao Gao**, Haoyu Wang, Haiying Shen
Smartly handling renewable energy instability in supporting a cloud datacenter
IEEE International Parallel and Distributed Processing Symposium, 2020

ICCCN '20
2. **Jiechao Gao**, Haoyu Wang, Haiying Shen
Machine learning based workload prediction in cloud computing
International Conference on Computer Communications and Networks, 2020

BigData '19
1. **Jiechao Gao**, Haoyu Wang, Haiying Shen
Task failure prediction in cloud data centers using deep learning
IEEE International Conference on Big Data, 2019

Peer-reviewed Journal Papers

NPJ AI '26
30. **Jiechao Gao***, Yuangang Li*, Jie Wang, Michael Lepech, Yue Zhao, Brad Campbell
H-FedSN: Personalized Sparse Networks for Efficient and Accurate Hierarchical Federated Learning for IoT Applications
NPJ Artificial Intelligence, 2026

IEEE TAI '26 29. Gautam Siddharth Kashyap, Niharika Jain, Ebad Shabbir, Harsh Joshi, Usman Naseem, **Jiechao Gao**†
CLARITY: A Lightweight Multimodal Transformer for Harmful Content Detection
IEEE Transactions on Artificial Intelligence, 2026

AutoCon '26 28. Yuandong Pan, Mudan Wang, **Jiechao Gao**, Jie Wang, Michael D. Lepech, Ioannis Brilakis
LLM-enabled multi-agent framework for automated Scan-to-BIM and Scan-to-Graph reconstruction
Automation in Construction, 2026

AI for Eng. '25 27. Keyi Wang, Nikolaus Holzer, Ziyi Xia, Yupeng Cao, **Jiechao Gao**, Anwar Walid, Kairong Xiao, Xiao-Yang Liu Yanglet
FinRL Contests: Data-Driven Financial Reinforcement Learning Agents for Stock and Crypto Trading
Artificial Intelligence for Engineering, 2025

ACM TMIS '24 26. Xiao-Yang Liu, Rongyi Zhu, Daochen Zha, **Jiechao Gao**, Shan Zhong, Meikang Qiu
Differentially Private Low-Rank Adaptation of Large Language Model Using Federated Learning
ACM Transactions on Management Information Systems, 2024

EPSR '24 25. Aamir Nawaz, Hongtao Wang, Huiting Yang, Hammad Armghan, **Jiechao Gao**†
Risk-constrained probabilistic coordination in coupled transmission and distribution system
Electric Power Systems Research, 2024

Veh. Comm. '24 24. Mehdi Gheisari, Aminreza Karamoozian, **Jiechao Gao**, Hemn Barzan Abdalla, Shuja Ansari, Riaz Ullah Khan, Zhaoxi Fang
Accident reduction through a privacy-preserving method on top of a novel ontology for autonomous vehicles with the support of modular arithmetic
Vehicular Communications, 2024

IEEE Access '24 23. M. Mahad Malik, Abdullah Altamimi, Syed Ali Abbas Kazmi, Zafar Ali Khan, Muhammad Waleed Ansari, Kamran Mujahid, **Jiechao Gao**
A Full-Fledged, Multi-Agent System Representing the Architecture of Smart Cities by Balancing Energy With Optimal Electricity Forecasting, Integrating Individual Comfort, and Extracting Financial Gains
IEEE Access, 2024

TPDS '23 22. Haiying Shen, Haoyu Wang, **Jiechao Gao**, Rajkumar Buyya
An Instability-Resilient Renewable Energy Allocation System for a Cloud Datacenter
IEEE Transactions on Parallel and Distributed Systems, 2023

MLJ '23 21. Xiao-Yang Liu, Ziyi Xia, Hongyang Yang, **Jiechao Gao**, Daochen Zha, Ming Zhu, Christina Dan Wang, Zhaoran Wang, Jian Guo
Dynamic Datasets and Market Environments for Financial Reinforcement Learning
Machine Learning Journal, 2023

ESWA '23 20. Ahmad Almadhor, Rizwana Irfan, **Jiechao Gao**†, Nasir Saleem, Hafiz Tayyab Rauf, Seifedine Kadry
E2E-DASR: End-to-end deep learning-based dysarthric automatic speech recognition
Expert Systems with Applications, 2023

ASC '23 19. Hafiz Tayyab Rauf, **Jiechao Gao***, Ahmad Almadhor, Ali Haider, Yu-Dong Zhang†, Fadi Al-Turjman
Multi population-based chaotic differential evolution for multi-modal and multi-objective optimization problems
Applied Soft Computing, 2023

CAAI TIT '23 18. Nasir Saleem, **Jiechao Gao***, Rizwana Irfan, Ahmad Almadhor, Hafiz Tayyab Rauf, Yudong Zhang, Seifedine Kadry
DeepCNN: Spectro temporal feature representation for speech emotion recognition
CAAI Transactions on Intelligence Technology, 2023

JAIHC '23 17. Mehdi Gheisari, Amir Javadpour, **Jiechao Gao**†, Aaqif Afzaal Abbasi, Quoc-Viet Pham, Yang Liu
PPDMIT: A lightweight architecture for privacy-preserving data aggregation in the Internet of Things
Journal of Ambient Intelligence and Humanized Computing, 2023

JKE '23 16. Morshadul Hasan, Thuhid Noor, **Jiechao Gao**, Muhammad Usman, Mohammad Zoynul Abedin
Rural consumers financial literacy and access to FinTech services
Journal of the Knowledge Economy, 2023

TITS '22 15. Gunasekaran Manogaran, **Jiechao Gao***, Tu N Nguyen
Optimizing resource and service allocations for IoT-assisted intelligent transportation systems
IEEE Transactions on Intelligent Transportation Systems, 2022

TITS '22 14. **Jiechao Gao**, Gunasekaran Manogaran, Tu N Nguyen, Seifedine Kadry, Ching-Hsien Hsu, Priyan Malarvizhi Kumar
A vehicle-consensus information exchange scheme for traffic management in vehicular ad-hoc networks
IEEE Transactions on Intelligent Transportation Systems, 2022

IoT-J '22 13. Zeeshan Siddiqui, **Jiechao Gao**†, Muhammad Khurram Khan
An Improved Lightweight PUF-PKI Digital Certificate Authentication Scheme for the Internet of Things
IEEE Internet of Things Journal, 2022

JBHI '22 12. **Jiechao Gao**, Tu N Nguyen, Gunasekaran Manogaran, Ankit Chaudhary, Gai-Ge Wang
Redemptive resource sharing and allocation scheme for Internet of Things-assisted smart healthcare systems
IEEE Journal of Biomedical and Health Informatics, 2022

TNSE '22 11. Gunasekaran Manogaran, Tu N Nguyen, **Jiechao Gao**†, Priyan Malarvizhi Kumar
Deep learning-based service distribution model for wireless network assisted Internet of Everything
IEEE Transactions on Network Science and Engineering, 2022

ACM ToSN '22 10. Guimin Dong, Mingyue Tang, Zhiyuan Wang, **Jiechao Gao**, Sikun Guo, Lihua Cai, Robert Gutierrez, Bradford Campbell, Laura E Barnes, Mehdi Boukhechba
Graph Neural Networks in IoT: A Survey
ACM Transactions on Sensor Networks, 2022

TGCN '22 9. Tu N Nguyen, **Jiechao Gao**†, Gunasekaran Manogaran, R Dinesh Jackson Samuel, Mamoun Alazab
Transfer learning-aided collaborative computational method for intelligent transportation system applications
IEEE Transactions on Green Communications and Networking, 2022

KBS '22 8. **Jiechao Gao***†, Nasir Saleem*, Muhammad Irfan Khattak, Hafiz Tayyab Rauf, Seifedine Kadry, Muhammad Shafi
DeepResGRU: Residual gated recurrent neural network-augmented Kalman filtering for speech enhancement and recognition
Knowledge-Based Systems, 2022

IVC '22 7. Nasir Saleem, **Jiechao Gao***, Muhammad Irfan, Elena Verdu, Javier Parra Fuente
E2E-V2SResNet: Deep residual convolutional neural networks for end-to-end video driven speech synthesis
Image and Vision Computing, 2022

CEE '22 6. Muhammad Irfan Khattak, Nasir Saleem, **Jiechao Gao**, Elena Verdu, Javier Parra Fuente
Regularized sparse features for noisy speech enhancement using deep neural networks
Computers and Electrical Engineering, 2022

Agriculture '22 5. Yuanzhi Pan, Hua Jin, **Jiechao Gao**†, Hafiz Tayyab Rauf
Identification of Buffalo Breeds Using Self-Activated-Based Improved Convolutional Neural Networks
Agriculture, 2022

Neurocomp. '22 4. Adeel Ahmed, Khalid Saleem, Osman Khalid, **Jiechao Gao**†, Umer Rashid
Trust-aware denoising autoencoder with spatial-temporal activity for cross-domain personalized recommendations
Neurocomputing, 2022

JBHI '21 3. Priyan Malarvizhi Kumar, Choong Seon Hong, Fatemeh Afghah, Gunasekaran Manogaran, Keping Yu, Qiaozhi Hua, **Jiechao Gao**
Clouds proportionate medical data stream analytics for internet of things-based healthcare systems
IEEE Journal of Biomedical and Health Informatics, 2021

FGCS '21 2. Mehdi Gheisari, Hamid Esmaeili Najafabadi, Jafar A Alzubi, **Jiechao Gao**†, Guojun Wang, Aaqif Afzaal Abbasi, Aniello Castiglione
OBPP: An ontology-based framework for privacy-preserving in IoT-based smart city
Future Generation Computer Systems, 2021

TSC '20 1. **Jiechao Gao**, Haoyu Wang, Haiying Shen
Task failure prediction in cloud data centers using deep learning
IEEE Transactions on Services Computing, 2020

Peer-reviewed Workshop, Poster & Demo Papers

ICWSM W '26
11. Daniel Koo, **Jiechao Gao**, Yuandong Pan, Jie Wang, Michael D. Lepech
Perceiving Creativity in the Age of AI: How Labels, Beliefs, and Familiarity Shape Evaluations of AI-Generated and Human-Created Art
Workshop on Social Influence in the Era of LLMs (SocialLLM) at ICWSM, 2026

ICCPS '25 Poster
10. Tushar Routh, **Jiechao Gao**, Bradford Campbell
Data Efficient PV based Indoor Event Detection
ACM/IEEE International Conference on Cyber-Physical Systems, Poster Track, 2025

ACL '25 SRW
9. Ellen Yi-Ge, **Jiechao Gao**, Wei Han, Wei Zhu†
DRUM: Learning Demonstration Retriever for Large MUlti-modal Models
Annual Meeting of the Association for Computational Linguistics Student Research Workshop, 2025

ACL '25 SRW
8. Juyuan Zhang, **Jiechao Gao**, Wenwen Ouyang, Wei Zhu†, Hui Yi Leong
Time-LlaMA: Adapting Large Language Models for Time Series Modeling via Dynamic Low-rank Adaptation
Annual Meeting of the Association for Computational Linguistics Student Research Workshop, 2025

NeurIPS W '25
7. Chunlin Feng, Lijian Huang, Keyi Wang, Yupeng Cao, Ming Zhu, **Jiechao Gao**, Xiao-Yang Liu
Standard Market Environments for Financial Reinforcement Learning
NeurIPS 2025 Workshop on Generative AI in Finance, 2025

AAAI Bridge '23
6. Berend Jelmer Dirk Gort*, Xiao-Yang Liu*, Xinghang Sun, **Jiechao Gao**, Shuaiyu Chen, Christina Dan Wang.
Deep Reinforcement Learning for Cryptocurrency Trading: Practical Approach to Address Backtest Overfitting
AAAI Bridge Program on AI for Financial Services, 2023

SenSys '22 Poster
5. **Jiechao Gao***, Mingyue Tang*, Tianhao Wang, Bradford Campbell
Pfed-ldp: A personalized federated local differential privacy framework for iot sensing data
ACM Conference on Embedded Networked Sensor Systems, Poster Track, 2022

IPSN '22 Poster
4. **Jiechao Gao**, Wenpeng Wang, Bradford Campbell
Residential Energy Management System Using Personalized Federated Deep Reinforcement Learning
ACM/IEEE International Conference on Information Processing in Sensor Networks, Poster Track, 2022

NeurIPS '21 W
3. Xiao-Yang Liu, Jingyang Rui, **Jiechao Gao**, Liuqing Yang, Hongyang Yang, Zhaoran Wang, Christina Dan Wang, Jian Guo
FinRL-Meta: A Universe of Near-Real Market Environments for Data-Driven Deep Reinforcement Learning in Quantitative Finance
Conference on Neural Information Processing Systems, 2021

GLOBECOM W '21
2. M Poongodi, Mounir Hamdi, **Jiechao Gao**, Hafiz Tayyab Rauf
A Novel Security Mechanism of 6G for IMD using Authentication and Key Agreement Scheme
IEEE Global Communications Conference Workshops, 2021

SenSys W '21
1. **Jiechao Gao**, Wenpeng Wang, Zetian Liu, Md Fazlay Rabbi Masum Billah, Bradford Campbell
Decentralized federated learning framework for the neighborhood: a case study on residential building load forecasting
ACM Conference on Embedded Networked Sensor Systems, AIChallengeIoT Workshop, 2021

Peer-reviewed Book Chapters

Book '22
3. **Jiechao Gao**, Tiange Xie
DNA computing in cryptography
Advances in Computers (Elsevier), 2022

Book '20
2. Hongzhuo Qi, Yi Qin, **Jiechao Gao**
Cloud Computing and Big Data
Harbin Institute of Technology Press, 2020

Book '19
1. Jinglan Wu, Dongliang Lin, **Jiechao Gao**, Ye Xu
Introduction of Bigdata and Artificial Intelligence Applications
University of Electronic Science and Technology of China Press, 2019

Preprints

arXiv '26
4. PrivateHub: Contrastive Diffusion Model for Private Sensor-Intensive Environment Data Generation

arXiv '26
3. The Orchestration Gap: Why Process Automation Stalls in Operationally Complex Industries

arXiv '26 2. Clusters are All You Need: Pre-Training the Tsetlin Machine with Semantic Clusters from Language Models for Interpretability

Preprints.org '24 1. Artificial Intelligence-Aided Digital Twin Design: A Systematic Review and Future Directions

Grant & Proposal Experience

Contributed to funded research and proposal preparation as a graduate researcher / postdoc (not PI).

NSF: 2217071 Jul. 2022 - Sep. 2027
Collaborative Research: **PPoSS: LARGE: Co-designing Hardware, Software, and Algorithms to Enable Extreme-Scale Machine Learning Systems**
PI/co-PI(s): Aidong Zhang, Matthew Dwyer, Kevin Skadron, Bradford Campbell, Ashish Venkat

USDOT *under submission*
University Transportation Centers Program - Tier 1 Center
CIFE

Student Mentorship

20+ students have worked with me on research projects, many of whom moved on to PhD programs. Venues in parentheses are papers they first-authored or co-authored during our collaboration.

- Muran Yu: Stanford PhD (EMNLP'26 Ind.)
- Yulin Xu: UCI CS MS → USC CS PhD (KDD'25, ICML'26, SIGIR'26)
- Yuangang Li: USC CS MS → UCI CS PhD (BIBM'24, ICDM'24 W, AAAI'26 ×2, Findings of ACL'26, NeurIPS D&B'26, NPJ AI'26)
- Tiange Xie: Columbia U MS → IIE@CAS PhD (SciSec'24, ICASSP'26 ×2)
- Mingyue Tang: USC CS MS → UVA SE PhD → UIUC CS PhD (SenSys'22 Poster, ACM ToSN'22, ICCPS'25)
- Gautam Siddharth Kashyap: Jamia Hamdard → Macquarie University CS PhD (ICASSP'25 ×2, ASONAM'25, EMNLP'25 ×2, EACL'26 ×2, WWW'26, ACL'26, IEEE TAI'26)
- Berend Gort: Columbia U MLE → Universitat Politècnica de Catalunya CS PhD (AAAI Bridge'23)
- Kai Shu: USC CS MS (ACM MM'24, BIBM'24)
- Xinyuan Huang: U of Toronto CS MS → U of Toronto CS PhD (BigData'25 ×2, BIBM'25 ×2, ICASSP'26)
- Binrui Wu: Fudan University (KDD'25, ACM MM'25 Oral, NeurIPS'25, ICML'26)
- Siyuan Chen: University of Bristol (NeurIPS'25, ICLR'26)
- Haoran Ma: Harvard University (ICML'26, NeurIPS'26 ×3)
- Linxiao Li: University of Sydney (ICML'26, NeurIPS'26 ×2)
- Haochen Sui: University of Michigan (ACM MM'25 ×2, NeurIPS'26 ×2)
- Qi Sun: University of Melbourne (SIGIR'26)
- Yidong Wu: Imperial College London (NeurIPS'25)
- Ebad Shabbir: DSEU-Okhla → MBZUAI (EMNLP'25 ×2, EACL'26, ICASSP'26, ACL'26 ×2, IEEE TAI'26)
- Rafiq Ali: DSEU-Okhla → MBZUAI (ICASSP'25, EMNLP'25 ×2, EACL'26, ACL'26)
- Harsh Joshi: Bharati Vidyapeeth's College of Engineering → UC San Diego MS (ICASSP'25, EACL'26, IEEE TAI'26)
- Daniel Koo: The Nueva School (high school) (ICWSM'26 W)

Talks and Lectures

The Chinese University of Hong Kong, Shenzhen	*Translational AI Systems: What Has to Be True Before Anyone Can Rely on What a Model Produces*	Apr. 2026
EMNLP 2025 Industry Track, Suzhou	*Truth, Trust, and Trouble: Medical AI on the Edge* (Best Paper Nomination)	Nov. 2025
ACM Multimedia 2025, Dublin	*Prototype-Guided Representation Projection for Multi-Domain Multi-Task Recommendation* (Oral, top 4%)	Oct. 2025
Stanford University	*Private and Accurate IoT Applications*	May. 2025
ICCPS 2025, Irvine	*Atlas: Ensuring Accuracy for Privacy-Preserving Federated IoT Applications*	May. 2025
Rutgers Efficient AI Seminar	*Federated Learning Enabled IoT Systems*	May. 2024
ICPP 2023, Salt Lake City	*PFDRL: Personalized Federated Deep Reinforcement Learning for Residential Energy Management*	Aug. 2023
CHIL 2023, New York	*SRDA: Mobile Sensing based Fluid Overload Detection for End Stage Kidney Disease Patients* (Oral)	Jun. 2023
Commonwealth Cyber Initiative Symposium	*PFed-LDP: A Personalized Federated Local Differential Privacy Framework for IoT Sensing Data*	Apr. 2023
Beijing Institute of Technology	*Towards Accurate and Private Cloud-IoT Applications*	Feb. 2023
Jilin University	*Towards Accurate and Private Cloud-IoT Applications*	Feb. 2023
Beijing University of Technology	*Towards Accurate and Private Cloud-IoT Applications*	Feb. 2023
UVA ML Seminar	*Federated Learning with Privacy*	Aug. 2022
UVA Link Lab	*Towards Accurate and Private IoT Applications*	Feb. 2022
IPDPS 2020 (virtual)	*Smartly Handling Renewable Energy Instability in Supporting a Cloud Datacenter*	May. 2020
Frees Fund	*How Neural Networks and Deep Learning Work in the Financial Industry*	Aug. 2018

Teaching Experience

Stanford University — Stanford, CA

Guest Lecture — Winter, 2026

Stanford University CEE 221

University of Virginia — Charlottesville, VA

Teaching Assistant — Fall 2020

Computer Networks (Prof. Brad Campbell)

Teaching Assistant (helping on course topics) — Spring 2020

Advanced Computer Networks (Prof. Yixin Sun)

Teaching Assistant — Fall 2019

Principles of Modeling for Cyber-Physical Systems (Prof. Madhur Behl)

Columbia University — New York City, NY

Teaching Assistant — Fall 2017

Reinforcement Learning (Prof. Chong Li)

Services

Conference/Workshop/Competition Organizing Committee

- ASONAM Multidisciplinary Track Co-Chair — 2026
- FinRL Contest ICAIF — 2023, 2024, 2025
- UbiComp/ISWC Workshop on Design with Autistic Children — 2024
- SecureComm Workshop on S/P-IoT — 2022

Area Chair and/or (Senior) Program Committee/Reviewer for Conferences and Workshops

- IJCAI 2025 (SPC)
- AAAI-AIA 2025 (SPC)
- IJCNN 2025, 2026 (AC)
- EACL Industry Track 2026 (AC)
- IEEE IDS 2025, 2026 (Track Chair)
- NeurIPS 2022, 2023, 2024, 2025, 2026
- EMNLP Industry 2026
- SIGIR 2026
- BMVC 2026
- ICML 2025, 2026
- ACL ARR 2026
- KDD 2023, 2024, 2025, 2026
- AAAI 2024, 2025, 2026
- SDM 2024

- ML4H 2023, 2024
- UbiComp/IMWUT 2025
- ICLR 2025, 2026
- MLHC 2025
- ICASSP 2025, 2026 (AC)
- CVPR 2025, 2026
- ICCV 2025
- ICCPS 2025
- ACM MM 2025, 2026
- DSNE 2023–2026
- SecureComm 2022
- 2nd QTNML@NeurIPS 2022
- DCAA@AAAI 2023
- MILET@SIGKDD 2023
- AI4TS@IJCAI 2023
- Sensys Workshop on IoT Datasets 2024
- AI4TS@SDM 2024

Journal Reviewer

- Social Network Analysis and Mining (Associate Editor)
- IEEE Transactions on Systems, Man and Cybernetics: Systems
- Journal of Machine Learning Research (JMLR)
- IEEE Internet of Things Journal (IoT-J)
- IEEE Transactions on Intelligent Transportation Systems (TITS)
- IEEE Journal of Biomedical and Health Informatics (JBHI)
- IEEE Transactions on Cloud Computing (TCC)
- ACM Transactions on Sensor Networks (ToSN)
- Journal of Industrial Information Integration
- ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP)
- Financial Innovation
- ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)
- Expert Systems with Applications (ESWA)
- Cluster Computing
- IEEE Sensors Journal
- IEEE Access
- Soft Computing
- Journal of Ambient Intelligence and Humanized Computing
- Multimedia Tools and Applications

References

Prof. Bradford Campbell
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Prof. Michael Lepech
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Department of Civil and Environmental Engineering
Stanford University
Reference email: mlepech@stanford.edu

Prof. Jie Wang
Executive Director (Postdoc Advisor)
Center for Sustainable Development and Global Competitiveness
Department of Civil and Environmental Engineering
Stanford University
Reference email: jiewang@stanford.edu